INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-3

GUEST ANALYSIS MODULE

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PROJECT NAME: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

Step-1 : Creation of TblGuests Table

A table named **TblGuests** was created in Power BI and three columns were added: **RecencyScore**, **FrequencyScore**, and **SpendScore**.

```
TBI_Guests =
SUMMARIZE(
   'TBI_Bookings',
   'TBI_Bookings'[CustomerID],
   "FirstStay",MIN('TBI_Bookings'[CheckInDate]),
   "LastStay",MAX('TBI_Bookings'[CheckOutDate])
)
```

These columns are based on **RFM analysis**:

- **RecencyScore** – shows how recently a guest made a booking

```
RecencyScore =
DATEDIFF(
   CALCULATE(
      MAX('TBI_Bookings'[CheckOutDate]),
      FILTER('TBI_Bookings','TBI_Bookings'[CustomerID] = 'TBI_Guests'[CustomerID])
   ),
   TODAY(),DAY
)
```

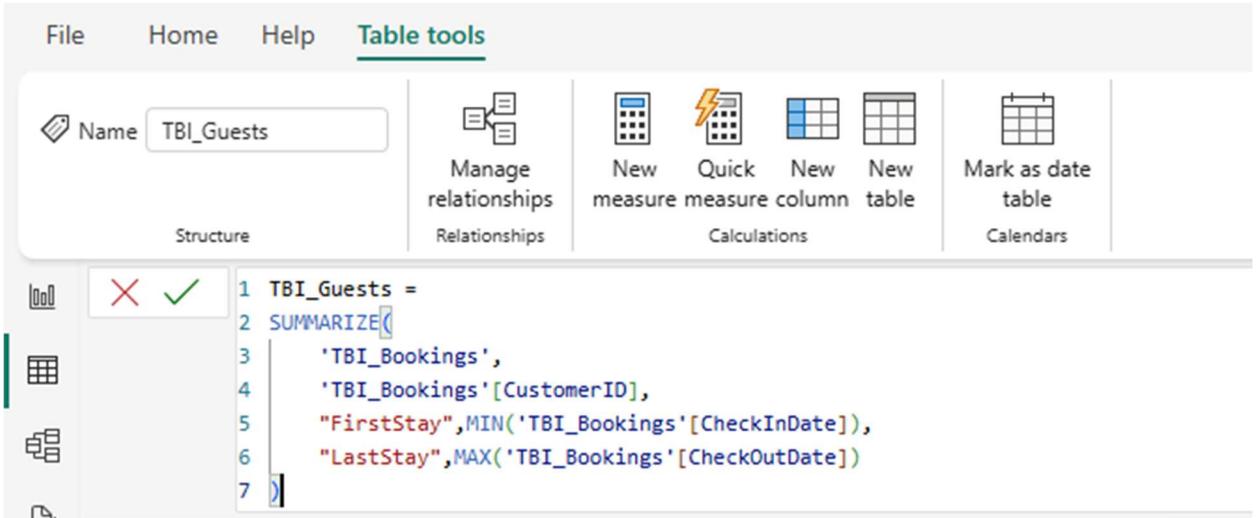
- **FrequencyScore** – shows how often a guest makes bookings

```
FrequencyScore =
CALCULATE(
   COUNTROWS('TBI_Bookings'),
   FILTER('TBI_Bookings','TBI_Bookings'[CustomerID] = 'TBI_Guests'[CustomerID])
)
```

- **SpendScore** – shows how much a guest spends

```
SpendScore =
CALCULATE(
    SUM(TBI_Bookings[DayRevenue]),
    FILTER('TBI_Bookings','TBI_Bookings'[CustomerID] = 'TBI_Guests'[CustomerID])
)
```

These scores help in identifying customer behavior and segmenting guests for better analysis and decision-making.



CustomerID	FirstStay	LastStay	FrequencyScore	SpendScore	RecencyScore
C2587	10/25/2023	3/20/2025	4	23800	366
C3709	7/25/2025	7/28/2025	1	12000	236
C2700	5/28/2025	5/29/2025	1	2500	296
C3158	10/7/2023	1/4/2026	4	16300	76
C2101	6/30/2024	2/5/2026	3	31000	44
C4894	10/11/2024	1/28/2026	2	15000	52
C2753	10/24/2024	9/5/2025	4	19000	197
C1997	5/1/2024	6/15/2025	5	41000	279
C4958	9/27/2023	3/26/2025	3	21000	360
C2098	3/26/2024	10/4/2025	6	37300	168
C456	5/4/2024	5/9/2024	1	7000	681
C3708	1/20/2024	1/20/2026	5	22300	60
C2938	3/21/2024	8/21/2025	7	55000	212
C2976	12/13/2023	10/20/2025	4	43500	152
C4811	5/5/2024	1/28/2026	7	43800	52
C742	9/30/2023	3/25/2025	3	19000	361
C3396	6/23/2024	11/8/2025	5	28800	133

Step-2: Updating Bookings Table with Additional Columns

In Power BI, the data was refreshed and the remaining dataset was loaded successfully. After that, additional columns were added to the existing **Bookings** table:

- **SourceType**

```
SourceType =
SWITCH(
   TRUE(),
   Booking[BookingChannel] = "Website", "Direct",
   Booking[BookingChannel] = "Mobile App", "Direct",
   Booking[BookingChannel] = "Call Center", "Direct",
   Booking[BookingChannel] = "Travel Agent", "Agent",
   "Other"
)
```

- **GuestType**

```
GuestType =
SWITCH(TRUE(),
   [purpose] = "Business" || [BookingChannel] = "Corporate", "Business",
   [purpose] = "Vacation" && [Stay Type] = "Extended Stay", "Family",
   [purpose] = "Holiday" && [Stay Type] = "Extended Stay", "Family",
   [purpose] = "Conference" || [Stay Type] = "Extended Stay", "Corporate","Solo"
)
```

- **RoomRate**

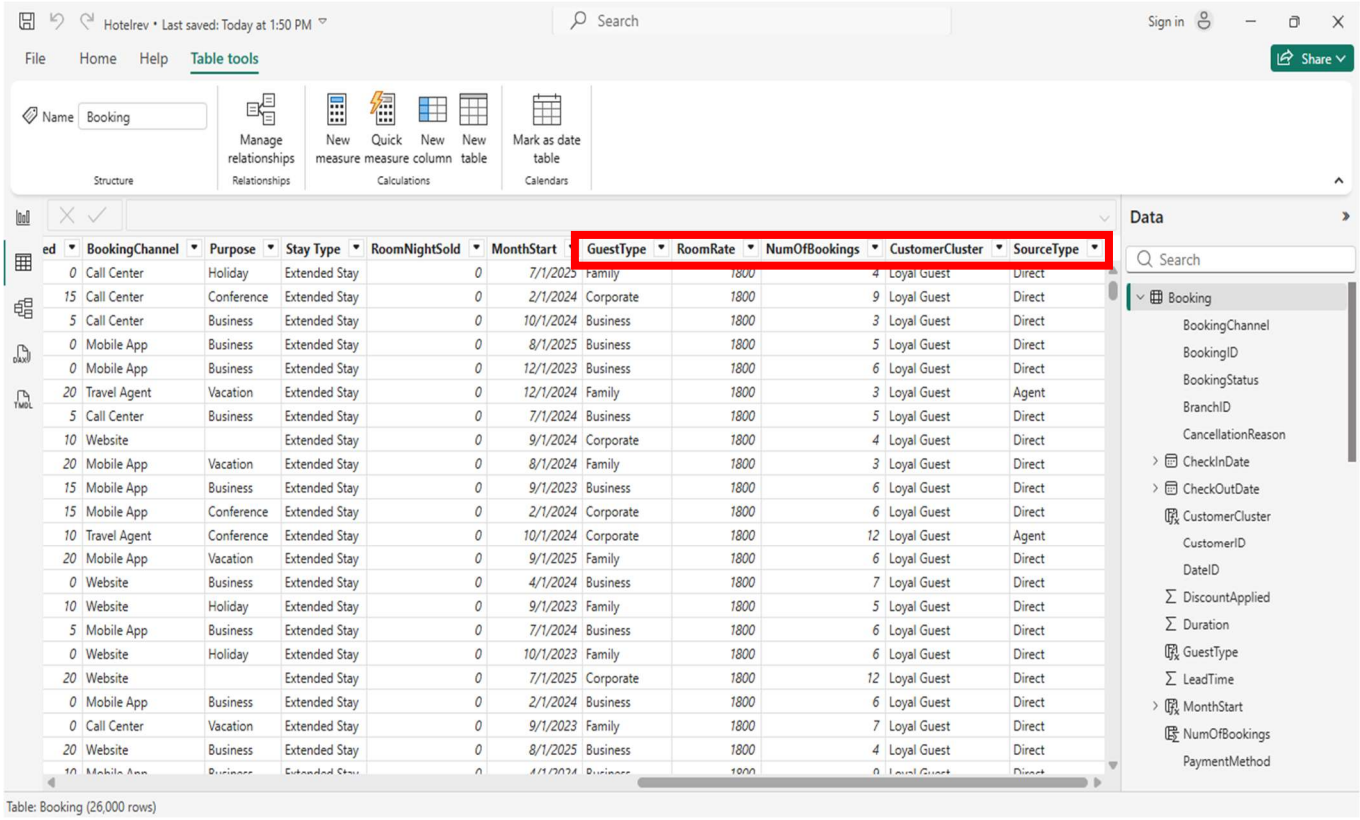
```
RoomRate = RELATED(RoomType[Price])
```

- **NumOfBookings**

```
NumOfBookings =
CALCULATE(
   COUNT(Booking[BookingID]),
   ALLEXCEPT(Booking,Booking[CustomerID])
)
```

- **CustomerCluster**

```
CustomerCluster =
SWITCH(TRUE(),
   [NumOfBookings]=1 && [Revenue]<5000,"First-Timer",
   [NumOfBookings]>=3 && [Revenue]<15000,"Loyal Guest",
   [Revenue]>= 15000,"High Spender","Other")
```



ed	BookingChannel	Purpose	Stay Type	RoomNightSold	MonthStart	GuestType	RoomRate	NumOfBookings	CustomerCluster	SourceType
0	Call Center	Holiday	Extended Stay	0	7/1/2025	Family	1800	4	Loyal Guest	Direct
15	Call Center	Conference	Extended Stay	0	2/1/2024	Corporate	1800	9	Loyal Guest	Direct
5	Call Center	Business	Extended Stay	0	10/1/2024	Business	1800	3	Loyal Guest	Direct
0	Mobile App	Business	Extended Stay	0	8/1/2025	Business	1800	5	Loyal Guest	Direct
0	Mobile App	Business	Extended Stay	0	12/1/2023	Business	1800	6	Loyal Guest	Direct
20	Travel Agent	Vacation	Extended Stay	0	12/1/2024	Family	1800	3	Loyal Guest	Agent
5	Call Center	Business	Extended Stay	0	7/1/2024	Business	1800	5	Loyal Guest	Direct
10	Website		Extended Stay	0	9/1/2024	Corporate	1800	4	Loyal Guest	Direct
20	Mobile App	Vacation	Extended Stay	0	8/1/2024	Family	1800	3	Loyal Guest	Direct
15	Mobile App	Business	Extended Stay	0	9/1/2023	Business	1800	6	Loyal Guest	Direct
15	Mobile App	Conference	Extended Stay	0	2/1/2024	Corporate	1800	6	Loyal Guest	Direct
10	Travel Agent	Conference	Extended Stay	0	10/1/2024	Corporate	1800	12	Loyal Guest	Agent
20	Mobile App	Vacation	Extended Stay	0	9/1/2025	Family	1800	6	Loyal Guest	Direct
0	Website	Business	Extended Stay	0	4/1/2024	Business	1800	7	Loyal Guest	Direct
10	Website	Holiday	Extended Stay	0	9/1/2023	Family	1800	5	Loyal Guest	Direct
5	Mobile App	Business	Extended Stay	0	7/1/2024	Business	1800	6	Loyal Guest	Direct
0	Website	Holiday	Extended Stay	0	10/1/2023	Family	1800	6	Loyal Guest	Direct
20	Website		Extended Stay	0	7/1/2025	Corporate	1800	12	Loyal Guest	Direct
0	Mobile App	Business	Extended Stay	0	2/1/2024	Business	1800	6	Loyal Guest	Direct
0	Call Center	Vacation	Extended Stay	0	9/1/2023	Family	1800	7	Loyal Guest	Direct
20	Website	Business	Extended Stay	0	8/1/2025	Business	1800	4	Loyal Guest	Direct

Step-3: Visualization Steps in Power BI

Step 1: Stacked Column Chart (Nationality vs Booking Count)

- Open Power BI
- Choose *Stacked Column Chart*
- Add Nationality to X-axis
- Add Guest Type to Legend
- Add BookingID to Values and set it to *Count*

Step 2: Clustered Column Chart (Booking Channel vs Guest Type)

- Select *Clustered Column Chart*
- Place Booking Channel on X-axis
- Place Guest Type in Legend
- Add BookingID to Values → set as *Count*

Step 3: Column Chart (Average Duration by Guest Type)

- Choose *Column Chart*
- Drag Guest Type to X-axis

- Drag Duration to Values → set as *Average*

Step 4: Pie Chart (Booking Status)

- Select *Pie Chart*
- Add Booking Status to Legend
- Add BookingID to Values → set as *Count*

Step 5: Pie Chart (Payment Method – Booking Count)

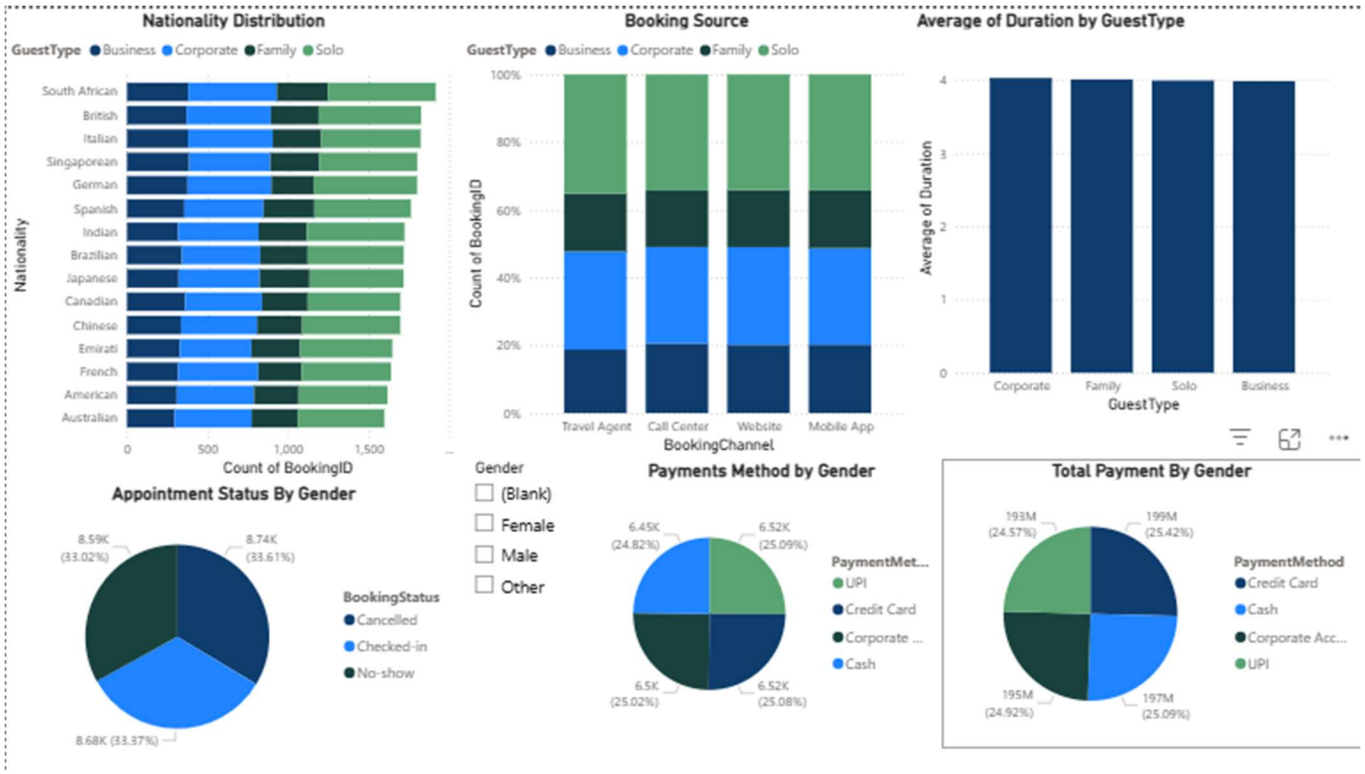
- Choose *Pie Chart*
- Drag Payment Method to Legend
- Drag BookingID to Values → set as *Count*

Step 6: Pie Chart (Payment Method – Revenue)

- Select *Pie Chart*
- Add Payment Method to Legend
- Add Revenue to Values → set as *Sum*

Step 7: Applying Filter (Gender)

- Insert a *Slicer* visual
- Add Gender field to slicer
- Select required options: Male, Female, Others



Step-4: Customer Analysis Visualizations

Scatter Plot Creation (Customer Analysis)

Step 1: Select Scatter Chart

- Open Power BI and switch to *Report View*
- Choose *Scatter Chart* from Visualizations pane

Step 2: Add Required Fields

- Place RecencyScore on X-axis
- Place FrequencyScore on Y-axis
- Place SpendScore in Size

Step 3: Set Aggregation

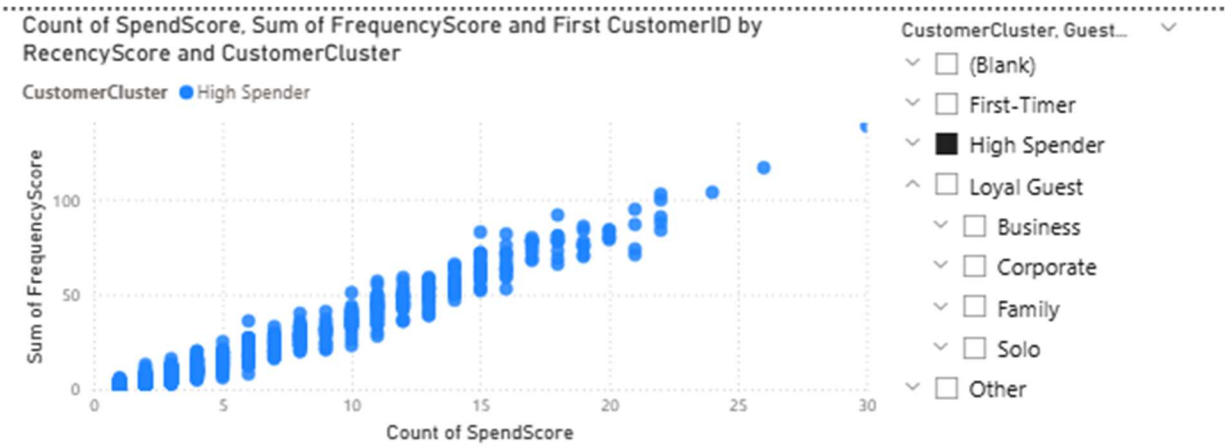
- Set SpendScore as *Count*
- Set FrequencyScore as *Sum*

Step 4: Add Legend

- Drag CustomerCluster into Legend

Step 5: Apply Filters (Slicers)

- Insert *Slicer* and add GuestType
- Insert another *Slicer* and add CustomerCluster
- Insert one more *Slicer* and add SourceType
- Select values to filter the scatter plot



Step-5: Map Visualization (Nationality vs Customer Cluster)

Step 1: Create Map

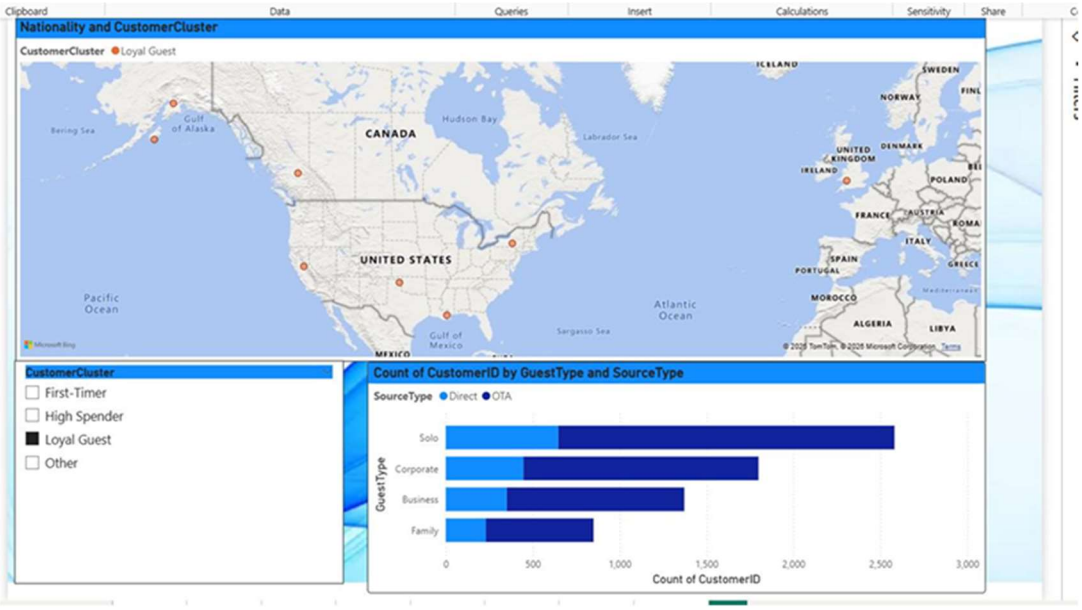
- Select *Map Visual* in Report View
- Add Nationality to Location
- Add CustomerCluster to Legend
- Add CustomerID to Size → set as *Count*

Stacked Column Chart (Customer Count Analysis)

Step 1: Create Chart

- Select *Stacked Column Chart*
- Add GuestType to X-axis
- Add SourceType to Legend
- Add CustomerID to Values → set as *Count*

Apply Slicer (Customer Cluster)

Step 1: Add Filter

- Insert *Slicer* visual
- Drag CustomerCluster into slicer
- Choose required values to filter all visuals